Transfer Functions — Theory, Derivation, and Applications

Objective

This note introduces transfer functions as the primary modeling tool for linear time-invariant (LTI) systems.
The focus is on:

• Definition and derivation
• Interpretation of poles and zeros
• Time-domain and frequency-domain meaning
• Application to control systems

The goal is to establish a clear analytical foundation for feedback and controller design.

Definition of Transfer Function

For an LTI system under zero initial conditions, the transfer function is defined as:

$$G(s) = \frac{Y(s)}{X(s)}$$

where:

$\boldsymbol{X(s)}$is the Laplace transform of the input

$\boldsymbol{Y(s)}$is the Laplace transform of the output

$$\boldsymbol{s = \sigma + j\omega}$$

Zero initial conditions are assumed so that the transfer function depends only on system dynamics.

Derivation from Differential Equation

Example: RC circuit

$$RC\frac{dv_o(t)}{dt} + v_o(t) = v_i(t)$$

Taking Laplace transform:

$$RC\, sV_o(s) + V_o(s) = V_i(s)$$

Solving for the ratio:

$$\frac{V_o(s)}{V_i(s)} = \frac{1}{RCs + 1}$$

This is the transfer function of a first-order low-pass filter.

General Transfer Function Form

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0}$$

Numerator polynomial defines zeros

Denominator polynomial defines poles

System order = highest power of s in denominator

Poles and Zeros

Poles are solutions of:

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0 = 0$$

Zeros are solutions of:

$$b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0 = 0$$

Stability condition:

$$\mathrm{Re}(s_i) < 0 \quad \forall \text{ poles}$$

If any pole has positive real part, the system is unstable.

First-Order System Example

$$G(s) = \frac{1}{RCs + 1}$$

Time constant:

$$\tau = RC$$

Pole location:

$$s = -\frac{1}{RC}$$

Step Response

For a step input:

$$v_{in}(t) = V_0 u(t)$$

Output:

$$v_o(t) = V_0\left(1 - e^{-t/RC}\right)$$

Key properties:

Exponential rise

Settles to $\boldsymbol{V_0}$

63% rise at $\boldsymbol{t = RC}$

Transfer Functions in Control Systems

PI Controller:

$$G_{PI}(s) = K_p + \frac{K_i}{s}$$

Closed-loop system:

$$G_{cl}(s) = \frac{G(s)}{1 + G(s)H(s)}$$

Block Diagram Operations

Series:

$$G_{eq}(s) = G_1(s)G_2(s)$$

Parallel:

$$G_{eq}(s) = G_1(s) + G_2(s)$$

Feedback:

$$G_{cl}(s) = \frac{G(s)}{1 + G(s)H(s)}$$

Frequency Response

Substitute:

$$s = j\omega$$

$$G(j\omega)$$

Used to determine:

- Magnitude response
- Phase response
- Bandwidth
- Gain margin
- Phase margin

Summary

• A transfer function represents the input–output behavior of an LTI system in the s-domain.
• It is derived from differential equations under zero initial conditions.
• Poles determine stability and dynamic behavior.
• Zeros shape transient and frequency response.
• Transfer functions form the foundation of feedback control analysis and controller design.